

Teacher Guide — senior / module-08-yoga

IKS Curriculum

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Overview

Module 8 — Yoga: Patañjali, Physiology, and the Modern Research Literature

Age band: Senior · **Grades:** 9–12 · **Duration:** 10 sessions × 45 min (with ~15 min movement)

Learning objectives

By the end of this module, students will be able to:

1. **Quote** Yoga Sūtras 1.2, 2.46, 2.49, and 2.50 in IAST with translation, and explain each.
2. **Recall and apply** the eight limbs (*aṣṭāṅga*) in their classical Sāṅkhya-Yoga context.
3. **Practise** the five foundational *āsanas* + Sūrya Namaskāra + counted *prāṇāyāma* safely, with awareness of contraindications.
4. **Articulate** the contemporary research on yoga's effects on the autonomic nervous system, including the polyvagal-theory framing (Stephen Porges).
5. **Distinguish** the *modern postural yoga* tradition (Krishnamacharya → Iyengar / Pattabhi Jois → present) from Patañjali's classical eight-limb system, and discuss what each preserves and modifies.
6. **Construct** a 5–6 page research-synthesis paper on a specific yoga-related question, integrating primary classical sources with peer-reviewed modern research.

NEP 2020 alignment

- §4.6 — Integration of IKS
- §4.31 — Physical fitness and well-being
- §11.8 — Multidisciplinary (philosophy + physiology + research methods)
- §4.27 — Indian cultural heritage in education

Prerequisites

- Familiarity with Middle-band Module 8 content (eight limbs, basic *āsanas*, counted breath).

- Medical opt-out form on file.
- Comfort reading Sanskrit verses with provided transliteration and translation.

Key vocabulary (Senior additions)

- *citta-vṛtti-nirodha* — settling the fluctuations of consciousness (YS 1.2)
- *prayatna-śaithilya, ananta-samāpatti* — the dual conditions for āsana (YS 2.47)
- *Patañjali* — author/compiler of the *Yoga Sūtras*
- *Krishnamacharya* — 20th-century teacher; root of modern postural yoga
- *vagus nerve, parasympathetic, polyvagal* — modern physiology terms

Materials checklist

- Yoga mats
- Printed extracts: YS 1.1–4, 2.46–50 with IAST + translation
- Printed extract: Porges (2009) “The polyvagal theory: new insights into adaptive reactions of the autonomic nervous system” (or teacher synopsis)
- Printed extract: Bryant (2009), *Yoga Sūtras of Patañjali*, commentary on YS 2.46–50

Day-by-day overview

Day	Title	Focus	Assessment
1	Hook: Three Definitions of Yoga	YS 1.2 + BG 2.50 + modern	Discussion
2	The Eight Limbs (Senior)	Full enumeration + classical sources	Limb quiz
3	Āsana — Patañjali’s Definition (YS 2.46)	Primary source close reading	Practical + reflection
4	Sūrya Namaskāra + Breath Co-ordination	Practice + history of modern SN	Practical observation
5	Mid-module Check + Prāṇāyāma Primary Source	YS 2.49–50 reading	Formative
6	Physiology + Polyvagal Theory	Porges reading	Reading response
7	Modern Postural Yoga — A Historical Note	Krishnamacharya, Iyengar, Pattabhi Jois	Discussion
8	Project Session 1	Choose topic + outline	Outline
9	Project Session 2	Draft + buddy	Draft

Day	Title	Focus	Assessment
		critique	
10	Final Assessment + Paper	Summative + paper	Summative + paper

Citation sources used

- **Bryant, E. F. (2009).** *The Yoga Sūtras of Patañjali*. North Point Press. Primary translation + commentary.
- **vidya-karana RAG** cached chunks (.rag-cache/): SB 3 (eight limbs purport), SB 4.8.44 (prāṇāyāma verse), BG Ch. 4 + 6 (breath sacrifice, dhyāna-yoga).
- **Porges, S. W. (2009).** “The polyvagal theory: new insights into adaptive reactions of the autonomic nervous system.” *Cleveland Clinic Journal of Medicine*, 76(Suppl 2), S86–S90.
- **Singleton, M. (2010).** *Yoga Body: The Origins of Modern Posture Practice*. Oxford University Press. — On the 20th-century history of postural yoga.

See sources.md for full list.

Project brief

5–6 page research synthesis paper on a specific yoga-related question. Choose from suggested topics or propose your own. Required: ≥1 primary classical source (YS or BG), ≥1 peer-reviewed modern research paper, ≥1 historical source (Singleton or equivalent). Full brief in assessments/project-brief.md.

Safety note

All Middle-band safety rules apply. At Senior band, *inversions are allowed only* if the school has a qualified yoga teacher and the student has demonstrated readiness over multiple sessions. **Default: no inversions.** Breath retention (kumbhaka) is permitted only with explicit teacher supervision and contraindication screening.

10-Day Lesson Plan

Module 8 (Senior) — 10-Day Lesson Plan

At a glance

Day	Title	File	Output
1	Three Definitions of Yoga	days/day-01-hook.md	Discussion
2	Eight Limbs (Senior)	days/day-02-eight-limbs.md	Limb quiz
3	Āsana — Patañjali (YS)	days/day-03-asana-	Practical + reflection

Day	Title	File	Output
	2.46)	patanjali.md	
4	Sūrya Namaskāra	days/day-04-sun-salutation.md	Practical observation
5	Prāṇāyāma + Primary Source	days/day-05-pranayama.md	Formative
6	Polyvagal Theory	days/day-06-polyvagal.md	Reading response
7	Modern Postural Yoga	days/day-07-modern-yoga.md	Discussion
8	Project Session 1	days/day-08-project-session-1.md	Outline
9	Project Session 2	days/day-09-project-session-2.md	Draft
10	Final	days/day-10-final-assessment.md	Summative + paper

Pacing notes

- Each session opens with ~10–12 min movement, then theory/discussion.
- Day 6 (Porges reading) is dense — budget 25 min for reading + discussion.
- Day 7 (modern postural yoga history) often surprises students; allow space for surprise.

What students leave with

- Direct primary-source quotations from the Yoga Sūtras.
- Practical safe foundation in five āsanās + Sūrya Namaskāra + counted breath.
- Understanding of polyvagal-theory framing for yoga's effects.
- Critical understanding that modern postural yoga has a specific (and recent) lineage.
- One research-synthesis paper.

Teacher Notes

Teacher Notes — Module 8 Yoga (Senior)

Posture

Senior students can hold the tension between (a) yoga as classical Indian philosophy (Patañjali, eight limbs, *citta-vṛtti-nirodha*) and (b) yoga as a 20th-century postural movement

(Krishnamacharya, Iyengar, modern wellness culture). Both are real. The module asks students to see both clearly without conflating them.

The honest framing: **modern postural yoga, as taught in schools and studios worldwide, is largely a 20th-century synthesis. It draws on classical Indian texts AND on European physical-culture movements that arrived in India during the colonial period.** Singleton's *Yoga Body* (2010) is the standard reference for this history. This is uncomfortable for some students; honour the discomfort while sticking to the historical record.

Common misconceptions at this band

1. **“Sūrya Namaskāra is ancient.”** It isn't — the modern 12-pose sequence was codified by the Raja of Aundh in the early 20th century. The Sanskrit names are old; the *sequence* is recent.
2. **“All yoga is Patañjali's yoga.”** Patañjali's *Yoga Sūtras* are one strand. The Bhagavad-gītā teaches three other “yogas” (karma, bhakti, jñāna). Modern postural yoga focuses on āsana — the 3rd of Patañjali's eight limbs — but actually does very little of the *philosophical* work of the other limbs.
3. **“Polyvagal theory proves yoga works.”** Porges's theory is *one* framework for understanding autonomic responses. It is well-regarded but not unchallenged in neuroscience. Cite it as *a* framework, not as proof.

On the Yoga Sūtra primary sources

Critical verses for this module: - **YS 1.2:** *yogaś citta-vṛtti-nirodhaḥ* — “Yoga is the settling of the fluctuations of consciousness.” - **YS 2.46:** *sthira-sukham āsanam* — “Posture should be steady and comfortable.” - **YS 2.47:** *prayatna-śaithilyānanta-samāpattibhyām* — “Through relaxation of effort and absorption into the infinite.” - **YS 2.49:** *tasmin sati śvāsa-praśvāsayor gati-vicchedaḥ prāṇāyāmaḥ* — “When that [āsana] is established, prāṇāyāma is the cessation of inhalation and exhalation.” - **YS 2.50:** *bāhyābhyantara-stambha-vṛttir deśa-kāla-saṅkhyābhiḥ paridṛṣṭo dīrgha-sūkṣmaḥ* — “Prāṇāyāma is the regulation of inhalation, exhalation, and retention in terms of space, time, and number; it becomes long and subtle.”

Bryant (2009) provides excellent commentary on these. Have it accessible.

On polyvagal theory

- Porges's polyvagal theory (2009) frames the autonomic nervous system as a three-tier system (ventral vagal “social engagement” / sympathetic “fight-flight” / dorsal vagal “freeze”).
- This nuances the older two-mode (sympathetic / parasympathetic) framing taught at Middle band.
- Yoga's combination of breath + movement + interoception engages the ventral vagal complex — the “safe and connected” mode.
- Note: polyvagal theory has critics (e.g. Grossman, Taylor 2007). Senior students can engage this debate.

Differentiation hints

- **Tier up:** Read additional Patañjali commentary (Bryant’s full notes on YS 2.46–50). Read Singleton’s *Yoga Body* introduction.
- **Tier down:** Focus on practical safety + 1–2 primary verses. The history can be supplementary.

Sensitivity notes

- Some students will have strong family traditions around yoga’s religious dimensions. Honour this. The classroom frame is “yoga as one of India’s intellectual and physical traditions, taught here in its philosophical-and-physiological aspects.” Religious / devotional aspects are honoured but not centred.
- Some students will resist anything that suggests yoga isn’t ancient and unbroken. The Singleton history shouldn’t be presented as debunking — it’s *additional* knowledge. Modern postural yoga is also a real Indian achievement; just a more recent one.

Project oversight

- 5–6 page research-synthesis paper. Specific topics encouraged over general ones.
- Strong topic example: “Does the counted-breath prāṇāyāma practice taught at Middle band have measurable effects on heart-rate variability? Review of recent literature.”
- Weak topic: “The benefits of yoga.” Push for specificity.

Safety at Senior band

Re-read the Middle-band safety rules. At Senior band: - Inversions allowed ONLY with qualified teacher present. - Breath retention (kumbhaka) ONLY with screening. - Default to natural breath unless specifically supervised.

Sources

Sources — Module 8 Yoga (Senior)

Primary classical sources

Source	Used in	Anchor verses
Patañjali, <i>Yoga Sūtras</i>. Translation: Bryant (2009)	Days 1, 3, 5	YS 1.2 (yoga definition); 2.46 (āsana definition); 2.47–48; 2.49–50 (prāṇāyāma); 2.29 (eight limbs)
Bhagavad-gītā. Translations vary	Days 1, 4	BG 2.50 (skill in action); 4.29 (breath sacrifice); 6.10–13 (dhyāna-yoga seat and posture)
Śrīmad-Bhāgavatam 4.8.44.	Day 5	Prāṇāyāma + gradual purification verse. In vidya-karana RAG cache.

Secondary scholarly sources

- **Bryant, E. F. (2009).** *The Yoga Sūtras of Patañjali*. North Point Press. — Modern annotated translation; comprehensive commentary integrating classical Vyāsa and modern scholarship.
- **Iyengar, B. K. S. (1966).** *Light on Yoga*. Schocken. — The most influential modern āsana reference. Use Chapters on tādāsana, vṛkṣāsana, Sūrya Namaskāra.
- **Singleton, M. (2010).** *Yoga Body: The Origins of Modern Posture Practice*. Oxford University Press. — **Required reading on Day 7.** Documents the 20th-century history of postural yoga; controversial but well-evidenced.

Modern research sources

- **Porges, S. W. (2009).** “The polyvagal theory: new insights into adaptive reactions of the autonomic nervous system.” *Cleveland Clinic Journal of Medicine*, 76(Suppl 2), S86–S90. — Required reading on Day 6.
- **Streeter, C. C., et al. (2010).** “Effects of yoga versus walking on mood, anxiety, and brain GABA levels.” *Journal of Alternative and Complementary Medicine*, 16(11), 1145–1152. — One example of contemporary research; useful for the project.
- **Pascoe, M. C., et al. (2017).** “Yoga, mindfulness-based stress reduction and stress-related physiological measures: A meta-analysis.” *Psychoneuroendocrinology*, 86, 152–168. — Meta-analysis; good source for project research.

Critical / counter-evidence sources

- **Grossman, P., & Taylor, E. W. (2007).** “Toward understanding respiratory sinus arrhythmia: Relations to cardiac vagal tone, evolution and biobehavioural functions.” *Biological Psychology*, 74(2), 263–285. — Critical of some polyvagal claims; useful for Senior students engaging the debate.
- **De Michelis, E. (2004).** *A History of Modern Yoga: Patañjali and Western Esotericism*. Continuum. — Earlier complement to Singleton.

On the Krishnamacharya-Iyengar-Pattabhi Jois lineage

- **Sjoman, N. E. (1996).** *The Yoga Tradition of the Mysore Palace*. Abhinav Publications. — On Krishnamacharya’s eclectic sources for modern postural yoga.

Citation style for student papers

- Classical sources: (YS 2.46) inline; full reference to Bryant translation in works-cited.
- Scholarly: (Singleton 2010, p. X) or (Porges 2009).
- All papers must include works-cited in MLA or Chicago.

What to tell students about the source mix

Three distinct kinds of source: 1. **Classical primary** (YS, BG, SB) — what the tradition says. 2. **Historical scholarship** (Singleton, De Michelis) — how the tradition evolved into modern form. 3. **Modern empirical research** (Porges, Streeter, Pascoe) — what we can measure about yoga’s effects today.

A strong paper engages all three. A weak paper cherry-picks one.

Day 1 — Three Definitions of Yoga

Module: Yoga (Senior) · **Time:** 45 min (10 min movement)

Learning objective

Students articulate three distinct definitions of *yoga* (Patañjali, Bhagavad-gītā, modern) and recognise their non-equivalence.

Materials

- Mats
- Printed: YS 1.2, BG 2.50, BG 6.10–13

Flow

Settle (3 min)

Sukhāsana. 5 breaths.

Core (15 min) — three definitions

1. **Patañjali (Yoga Sūtra 1.2):** > *yogaś citta-vṛtti-nirodhaḥ* > “Yoga is the settling of the fluctuations of consciousness.”
 - This is about the *mind*. Patañjali frames yoga as a discipline aimed at quieting mental activity.
2. **Bhagavad-gītā (2.50):** > *yogaḥ karmasu kauśalam* > “Yoga is skill in action.”
 - This is about *action*. Krishna frames yoga as a way of acting without attachment to results.
3. **Modern postural yoga (any contemporary text):** > Yoga is a system of physical postures and breath techniques for health, flexibility, and stress reduction.
 - This is about the *body*. The modern frame is the one most people in 2026 encounter first.
4. **Are these the same?** Open question. Discuss:
 - Patañjali’s eight limbs include āsana (#3) but most limbs are mental/ethical.
 - The Gītā’s “skill in action” can be done without any āsana at all.
 - Modern postural yoga focuses on what classical Patañjali treated as one *limb*, not the whole.

Practice (12 min) — tādāsana + brief vṛkṣāsana

- 3 holds of tādāsana (30 sec each).
- 2 holds vṛkṣāsana per side.

- Settle in sukhāsana for 60 sec.

Discussion (10 min)

- “Which of the three definitions feels most accurate to YOUR experience of yoga?”
- “Are they compatible? Do they conflict?”

Wrap-up (5 min)

- Preview Day 2: eight limbs in their classical Sāṅkhya-Yoga context.

Differentiation

- **Tier up:** Read Bryant’s introduction to the *Yoga Sūtras* before Day 2. Bring 1 question.
- **Tier down:** Focus on memorising the two Sanskrit definitions (YS 1.2 and BG 2.50).

Teacher notes

- Some students will resist the framing “modern postural yoga is recent.” Save the deep history for Day 7; today just plant the three definitions.

Citation(s) used

- YS 1.2 — Bryant (2009)
- BG 2.50 — multiple translations
- vidya-karana RAG cache day-01-hook .json provides supporting BG and SB references

Day 2 — The Eight Limbs in Their Classical Context

Module: Yoga (Senior) · **Time:** 45 min (10 min movement)

Learning objective

Students enumerate the eight limbs with Patañjali’s classical definitions and locate the limbs within the Sāṅkhya-Yoga framework.

Materials

- Mats
- Eight-limbs wall chart
- YS 2.29 + 2.30, 2.32, 2.46, 2.49, 3.1, 3.2, 3.3 (key sūtras for each limb)

Flow

Settle (3 min)

Core (20 min)

1. **Read YS 2.29:** > *yama-niyamāsana-prāṇāyāma-pratyāhāra-dhāraṇā-dhyāna-samādhayo ’ṣṭāv aṅgāni* > “Restraints, observances, posture, breath regulation, sense-

withdrawal, focused attention, sustained meditation, absorption — these are the eight limbs.”

2. **Walk through each limb with its key sūtra:**

#	Limb	Sanskrit gloss	Source verse
1	<i>yama</i>	restraints (ahiṃsā, satya, asteya, brahmacarya, aparigraha)	YS 2.30
2	<i>niyama</i>	observances (śauca, santoṣa, tapas, svādhyāya, īśvara-praṇidhāna)	YS 2.32
3	<i>āsana</i>	“sthira-sukham” — steady and comfortable	YS 2.46
4	<i>prāṇāyāma</i>	breath regulation; cessation of gati	YS 2.49
5	<i>pratyāhāra</i>	sense-withdrawal	YS 2.54
6	<i>dhāraṇā</i>	focused attention	YS 3.1
7	<i>dhyāna</i>	sustained meditation	YS 3.2
8	<i>samādhi</i>	absorption	YS 3.3

3. **Note the proportions.** Patañjali’s *Yoga Sūtras* total ~196 sūtras across 4 chapters. Of these:

- Ethics + observances (yama, niyama): ~16 sūtras
- Āsana: ~3 sūtras (just YS 2.46–48)
- Prāṇāyāma: ~5 sūtras (YS 2.49–53)
- Pratyāhāra: ~2 sūtras
- Dhāraṇā + dhyāna + samādhi: most of Chapter 3 (~50 sūtras)

Modern postural yoga focuses on āsana — to which Patañjali devotes 3 sūtras out of 196.

Practice (12 min)

- Tāḍāsana → vṛkṣāsana → baddha-koṇāsana → balāsana → sukhāsana.

Limb quiz (5 min)

- 8 marks — name all 8 limbs in Sanskrit with English meaning.

Wrap-up (5 min)

- Preview Day 3: “Tomorrow we read YS 2.46 closely — Patañjali’s definition of āsana.”

Differentiation

- **Tier up:** Memorise the 5 *yamas* and 5 *niyamas* in Sanskrit.
- **Tier down:** Memorise the 8 limbs only.

Teacher notes

- The “proportions” observation is a powerful teaching moment. Most students assume yoga = āsana. The 3-out-of-196 statistic reframes this.

Citation(s) used

- Patañjali YS 2.29, 2.30, 2.32, 2.46, 2.49 — Bryant (2009).
- vidya-karana RAG cache day-02-eightlimbs.json.

Day 3 — Āsana: Patañjali’s Definition (YS 2.46)

Module: Yoga (Senior) · **Time:** 45 min (15 min movement)

Learning objective

Students read YS 2.46–48 in IAST + translation, identify the dual condition for āsana (steady AND comfortable), and reflect on whether modern āsana practice meets Patañjali’s criterion.

Materials

- Mats
- Printed YS 2.46–48 with Bryant’s commentary
- Practice space

Flow

Settle (3 min)

Reading (15 min)

1. **YS 2.46:** > *sthira-sukham āsanam* > “Posture should be steady and comfortable.”
 - 3 words. 3 Sanskrit words. *sthira* = steady, firm. *sukha* = comfortable, easeful. *āsanam* = posture.
 - The dual condition: BOTH steady AND comfortable. Not steady-but-uncomfortable. Not comfortable-but-collapsing.
2. **YS 2.47:** > *prayatna-śaithilyānanta-samāpattibhyām* > “[Sthira-sukham is achieved] through the relaxation of effort and absorption into the infinite.”
 - Two means to the dual condition: *prayatna-śaithilya* (relaxing effort) and *ananta-samāpatti* (absorption into the infinite / the limitless).
3. **YS 2.48:** > *tato dvandvānabhighātaḥ* > “Then [the practitioner is] no longer afflicted by the pairs of opposites.”
 - Result: equanimity in the face of dualities (hot/cold, comfort/discomfort, etc.).

Practice (15 min) — Test the definition

Hold tādāsana for 60 sec. Notice: - Are you *sthira* (steady)? - Are you *sukha* (comfortable)? - Are you both?

Now hold vṛkṣāsana (right). Same questions.

Then balāsana for 90 sec. Same questions.

Discussion (in pairs, 2 min): which pose best met Patañjali’s dual criterion for you, today?

Reflection writing (8 min)

5-minute quick-write: “Does modern postural yoga, with its emphasis on advanced and challenging poses, actually meet Patañjali’s criterion of steady AND comfortable? Argue your position in 4–5 sentences.”

Wrap-up (4 min)

- 2 students share their position.
- Preview Day 4: “Tomorrow — Sūrya Namaskāra. And a surprising historical question about its age.”

Differentiation

- **Tier up:** Read Bryant’s full commentary on YS 2.46–48. Bring 1 question.
- **Tier down:** Focus on memorising YS 2.46 in Sanskrit (3 words).

Teacher notes

- The dual criterion is genuinely counter-cultural to most modern yoga. Many studios push the “challenging hold” model. Patañjali doesn’t.
- The reflection writing often surfaces students’ own conflicted relationship with effort vs ease.

Citation(s) used

- Patañjali YS 2.46–48 — Bryant (2009), p. 282–286.

Day 4 — Sūrya Namaskāra: Practice and Historical Note

Module: Yoga (Senior) · **Time:** 45 min (20 min movement)

Learning objective

Students perform 2 rounds of Sūrya Namaskāra with breath co-ordination AND learn the surprising historical fact that the modern sequence is early-20th-century.

Materials

- Mats

- Wall poster of 12 SN poses
- Brief printed handout on SN history (teacher synopsis, 1 page)

Flow

Settle (3 min)

Core (10 min) — The history

- Sūrya Namaskāra in its modern 12-pose form was codified by the Raja of Aundh (Bhawanrao Shrinivasrao Pant Pratinidhi) in the early 20th century. He published *The Ten-Point Way to Health* in 1928, popularising the sequence.
- The individual āsanās have older roots. The sequence does not.
- This is one of many examples where modern postural yoga has a 20th-century pedigree, even when the Sanskrit names sound ancient.
- (We return to this on Day 7.)

Practice (20 min) — 2 rounds with breath

#	Pose	Breath
1	Praṇāmāsana	exhale
2	Hasta-uttānāsana	inhale
3	Uttānāsana	exhale
4	Aśva-saṅcalanāsana	inhale
5	Adho-mukha-śvānāsana	exhale
6	Aṣṭāṅga-namaskāra	(brief hold)
7	Bhujāṅgāsana	inhale
8	Adho-mukha-śvānāsana	exhale
9	Aśva-saṅcalanāsana	inhale
10	Uttānāsana	exhale
11	Hasta-uttānāsana	inhale
12	Praṇāmāsana	exhale

Round 1: slow, teacher-called. Round 2: at your own pace, breath-led.

Śavāsana for 2 min after.

Discussion (8 min)

- Did knowing the modern history change how the practice felt? Why or why not?
- Patañjali's *sthira-sukham* criterion — does Sūrya Namaskāra meet it?

Wrap-up (4 min)

- Preview Day 5: “Tomorrow — primary source reading on *prāṇāyāma*.”

Differentiation

- **Tier up:** Read about the Krishnamacharya lineage (Singleton ch. 8 brief).
- **Tier down:** 1 round at slower pace; focus on breath co-ordination.

Teacher notes

- The history surprises students. Some find it diminishing; others find it interesting. Both reactions are legitimate.
- Inversions in advanced SN variations are *not* taught here. Stick to the basic 12-pose sequence.

Citation(s) used

- Pant Pratinidhi (1928). *The Ten-Point Way to Health*.
- Singleton (2010). *Yoga Body*, ch. on SN.

Day 5 — Prāṇāyāma + Primary Source Reading

Module: Yoga (Senior) · **Time:** 45 min (15 min movement / breath)

Learning objective

Students read YS 2.49–50 in IAST + translation, practise counted breath safely, and complete the formative quiz.

Materials

- Mats
- Printed YS 2.49–50 with translation + commentary
- Printed SB 4.8.44 (from cache)
- Formative quiz Part B

Flow

Settle (3 min)

Primary source reading (12 min)

1. **YS 2.49:** > *tasmin sati śvāsa-praśvāsayor gati-vicchedaḥ prāṇāyāmaḥ* > “When that [āsana] is established, prāṇāyāma is the regulation [literally: severing of the flow] of inhalation and exhalation.”
 - Note: prāṇāyāma presupposes āsana. You don’t do prāṇāyāma in chaotic posture.
2. **YS 2.50:** > *bāhya-ābhyantara-stambha-vṛttir deśa-kāla-saṅkhyābhiḥ paridṛṣṭo dīrgha-sūkṣmaḥ* > “Prāṇāyāma has three modifications: outer (exhale), inner (inhale), and held (retained). It is observed by place, time, and number, and becomes long and subtle.”

- Three components: out-breath, in-breath, retention. Three measurements: place (in body), time (duration), number (count).
 - The goal: *dīrgha-sūkṣma* — long and subtle.
3. **SB 4.8.44:** > *prāṇāyāmena tṛ-vṛtā prāṇendriya-mano-malam śanair vyudasyābhidyāyen manasā guruṇā gurum* > “Through the threefold prāṇāyāma practice, one gradually purifies the breath, senses, and mind, and meditates.”

Practice (12 min) — Safe counted breath

- 5 min: natural-breath observation.
- 5 min: counted breath (4 in, 4 out). No retention.
- 2 min: śavāsana.

Senior-band caveat: With explicit teacher screening for contraindications (asthma, anxiety disorders), counted breath with brief retention (4-4-4-4: in 4, hold 4, out 4, hold 4) MAY be introduced. Default: skip retention.

Formative quiz Part B (15 min)

- 10 marks: limb identification, YS verse identification, safety question, short answer on YS 2.46 dual criterion, primary source reading question.

Wrap-up (3 min)

- Preview Day 6: “*Tomorrow — the modern research framing. Polyvagal theory.*”

Differentiation

- **Tier up:** Read Bryant’s commentary on YS 2.49–53. Note Vyāsa’s framing of subtler prāṇāyāma practices.
- **Tier down:** Stay with the natural breath; the YS verses can be re-read at home.

Teacher notes

- The “ascending from gross to subtle” arc of YS 2.49 → 2.50 → 2.51 is one of the most concise descriptions of prāṇāyāma in any tradition. Linger here.
- For safety: forceful prāṇāyāma is genuinely contraindicated for several conditions. Default position: natural breath + counted breath only.

Citation(s) used

- Patañjali YS 2.49–50 — Bryant (2009), p. 288–292.
- SB 4.8.44 — vidya-karana RAG cache day-06-pranayama.json.

Day 6 — Polyvagal Theory + Yoga

Module: Yoga (Senior) · **Time:** 45 min (10 min movement)

Learning objective

Students read Porges (2009) and articulate the polyvagal-theory framing of how yoga practices affect the autonomic nervous system.

Materials

- Mats
- Printed Porges (2009) or teacher synopsis (1–2 pages)
- Diagram of three-tier ANS (ventral vagal / sympathetic / dorsal vagal)

Flow

Settle (3 min)

Reading (15 min)

- Silent read of Porges (2009) extract.
- As they read, students note:
 - The three-tier ANS framework
 - What “ventral vagal” / “sympathetic” / “dorsal vagal” mean operationally
 - How yoga practices (specifically breath + posture + interoception) might engage the ventral vagal complex
 - One question they have

Core discussion (15 min)

1. Build the three-tier picture on the board:

Tier	State	Behaviour	Heart rate
Ventral vagal	Safe, social, connected	Calm engagement	Slow, variable (high HRV)
Sympathetic	Mobilised, alert	Fight or flight	Fast, low HRV
Dorsal vagal	Shutdown, conservation	Freeze, dissociation	Very slow

2. Yoga’s hypothesised effects (per Porges and others):

- Slow exhalation → ventral vagal activation.
- Steady āsana practice → interoceptive awareness → ventral vagal.
- Group yoga practice → social engagement → ventral vagal.

3. Limits and critiques.

- Porges’s theory has critics — Grossman & Taylor (2007) and others dispute specific claims about vagal physiology.
- The cleanest claim that survives: slow exhalation IS associated with parasympathetic activation (whether or not the polyvagal three-tier model is fully correct).

Practice (8 min) — Test the theory

- 1 min sukhāsana, natural breath, count heart rate (carotid pulse if comfortable).
- 5 min counted breath (4 in, 4 out).
- 1 min sukhāsana, count heart rate again.

Did anything change? (Most students notice a slight slowing. The N=1 anecdote doesn't prove anything, but it lets students *feel* the principle.)

Wrap-up (4 min)

- 2 students share their question from the reading.
- Preview Day 7: “*Tomorrow — modern postural yoga’s specific 20th-century lineage.*”

Differentiation

- **Tier up:** Read the Grossman & Taylor (2007) critique. Bring 1 specific objection to discuss.
- **Tier down:** Focus on the three-tier ANS framework; the polyvagal extensions can be supplementary.

Teacher notes

- The polyvagal framework is currently popular in wellness circles. It is *defensible* but not *settled*. Help students engage it as one framework, not as proof.

Citation(s) used

- Porges, S. W. (2009). “The polyvagal theory.” *Cleveland Clinic Journal of Medicine*.
- (Optional) Grossman & Taylor (2007). *Biological Psychology*.

Day 7 — Modern Postural Yoga: A Historical Note

Module: Yoga (Senior) · **Time:** 45 min (10 min movement)

Learning objective

Students engage Singleton’s (2010) thesis that modern postural yoga is largely a 20th-century synthesis, articulate one strength and one weakness of that thesis, and discuss what is at stake.

Materials

- Mats
- Printed extract from Singleton (2010), *Yoga Body* — Introduction + Ch. 1 (or teacher synopsis)
- Whiteboard

Flow

Settle (3 min)

Core (20 min)

1. **Singleton's thesis** (write on board): > Modern postural yoga, as taught in studios and schools worldwide today, is largely a 20th-century synthesis — drawing on classical Indian texts AND on European physical-culture movements that arrived in India during the colonial period (gymnastics, Swedish drill, bodybuilding, etc.).
2. **Specific examples Singleton gives:**
 - Sūrya Namaskāra in its modern form: early 20th c. (Raja of Aundh, 1928).
 - The strong-postural emphasis: Krishnamacharya (1888–1989), in Mysore Palace.
 - The flow + breath co-ordination: 20th-c. innovation, partly inspired by Western physical culture.
 - The international spread: Krishnamacharya's students Iyengar (1918–2014), Pattabhi Jois (1915–2009), Desikachar (1938–2016), TKV's daughter Indra Devi.
3. **What Singleton does NOT claim:**
 - Singleton does NOT say “yoga is fake.”
 - Singleton does NOT say “Patañjali is recent.”
 - Singleton does NOT say “Indians didn't invent yoga.”
 - Singleton says: *the modern postural form is recent, even though many of its component techniques are old.*
4. **Why this matters:**
 - For students: avoids the trap of either uncritical reverence or dismissive “it's all made-up.”
 - For policy: NEP §4.6 (“Integration of IKS”) works best when IKS is presented accurately, including the modern history.
 - For practice: knowing the history doesn't diminish the value. Modern postural yoga *works* — that's not in dispute.

Discussion (15 min)

- “What is *YOUR* reaction to Singleton's thesis? Does it diminish yoga, enrich your understanding, or feel like Western academic attack?”
- Hear from multiple students. Don't resolve the question.
- Add: “Singleton is a British scholar. Does that matter? Why or why not?”

Practice (5 min)

Brief Sūrya Namaskāra — 1 round. Same body, fresh awareness.

Wrap-up (2 min)

- Preview Day 8: “Tomorrow — *paper outlining.*”

Differentiation

- **Tier up:** Read De Michelis (2004) too. Compare and contrast with Singleton.
- **Tier down:** Focus on Singleton's central thesis; the further reading can wait.

Teacher notes

- This is the most politically loaded session of the module. Honour the discomfort some students will feel.
- The honest framing: *cultural achievements are not less valuable for being recent. Modern postural yoga is also a real Indian achievement; it's just a 20th-century one rather than an ancient one.*

Citation(s) used

- Singleton, M. (2010). *Yoga Body: The Origins of Modern Posture Practice*. Oxford University Press.
- De Michelis, E. (2004). *A History of Modern Yoga*. (Optional further reading.)

Day 8 — Project Session 1: Topic + Outline

Module: Yoga (Senior) · **Time:** 45 min

Learning objective

Students commit to a research-synthesis paper topic, draft a thesis paragraph, and produce an outline citing all 3 required source types.

Materials

- Project brief + rubric
- Drafting materials

Flow

Warm-up (5 min)

Topic selection (15 min)

Suggested topics: - “Does the counted-breath prāṇāyāma practice we learned have measurable effects on HRV? A literature review.” - “Patañjali defines āsana as ‘steady and comfortable’ (YS 2.46). Does modern postural yoga meet this criterion? Examine through 3 specific postures.” - “Singleton (2010) argues modern postural yoga is largely 20th century. Evaluate his evidence.” - “Compare Patañjali’s prāṇāyāma framework (YS 2.49–53) with the polyvagal theory’s account of breath effects.” - “Choose any one āsana and trace its history from classical text (if any) to modern practice. What does the history reveal?”

Each student commits to a topic. Discusses with teacher (60-sec rotation) — too broad? too narrow?

Outline (20 min)

Standard 5-6 page outline: - Intro with thesis - 3 body sections - Counter-argument - Conclusion
- Works cited (3+ sources, must include 1 classical primary + 1 historical scholarly + 1 modern research)

Wrap-up (5 min)

- Outline submitted at door.

Homework

- Begin drafting at home.

Differentiation

- **Tier up:** Add a 4th source type (e.g. an empirical study of an unrelated culture's somatic practice).
- **Tier down:** Narrow your topic further; one specific verse + one specific modern study is enough depth.

Teacher notes

- Strong topics are specific. Weak topics are broad. Push for specificity.

Day 9 — Project Session 2: Draft + Counter

Module: Yoga (Senior) · **Time:** 45 min

Learning objective

Each student reaches at least 2 pages drafted including a counter-argument and response.

Materials

- Drafts in progress

Flow

Warm-up (3 min)

- Brief: today's focus is the counter-argument.

Sprints (35 min)

- 15 min draft body
- 15 min buddy counter-argument workshop (cross-topic pairings recommended)
- 5 min incorporate

Wrap-up (7 min)

- 3 students share their best counter.

Homework

- Finish draft. Bring tomorrow.

Differentiation

- **Tier up:** Two distinct counters from different angles.
- **Tier down:** Detailed outline + 1 paragraph acceptable if a full draft is too much.

Teacher notes

- Common pattern: students avoid the counter because it feels like weakening their thesis. Reframe — a strong paper anticipates objections.

Day 10 — Summative + Paper Submission

Module: Yoga (Senior) · **Time:** 45 min

Learning objective

Students complete summative quiz and submit 5–6 page research-synthesis paper.

Materials

- Summative quiz
- Stopwatch

Flow

Quiz (35 min)

- 35 marks. MC, limb identification, primary-source reading, safety question, polyvagal-theory question, open response.

Paper submission (8 min)

- Print or upload. Teacher spot-checks.

Wrap-up (2 min)

- Reflection slip.
- Closing namaste.

Homework

- None. Module complete.

Differentiation

- **Tier up:** Begin reading next module's prerequisites.
- **Tier down:** Open-notebook quiz for documented learning needs.

Teacher notes

- Mark within 5 days.
- Research papers should be assessed on quality of synthesis, not on agreement with any particular position.

Module 8 (Senior) — Formative Quizzes

Part A — Day 2 (5 min)

1. (2 marks) Write the 8 limbs in Sanskrit, in order. 2. (1 mark) Of Patañjali's ~196 sūtras, how many are devoted directly to āsana? 3. (1 mark) Which limb does YS 2.46 (sthira-sukham āsanam) describe? 4. (1 mark) Name 2 of the 5 yamas. 5. (1 mark) Name 2 of the 5 niyamas.

Answer key

Q	Answer	Marks
1	yama, niyama, āsana, prāṇāyāma, pratyāhāra, dhāraṇā, dhyāna, samādhi	2
2	3 (YS 2.46–48)	1
3	āsana	1
4	Any 2 of: ahimsā, satya, asteya, brahmacarya, aparigraha	1
5	Any 2 of: śauca, santoṣa, tapas, svādhyāya, īśvara-praṇidhāna	1

Total: 6.

Part B — Day 5 (15 min)

1. (2 marks) Translate YS 2.46 word-by-word.
2. (2 marks) What is the dual condition Patañjali sets for āsana?
3. (2 marks) Translate the two operative words of YS 2.47: *prayatna-śaithilya* and *ananta-samāpatti*.
4. (2 marks) YS 2.49: in your own words, what does Patañjali say prāṇāyāma IS? (Note the role of āsana in the verse.)
5. (2 marks) Why does Senior-band practice limit retention (kumbhaka)?

Answer key

Q	Answer	Marks
1	sthira (steady, firm) + sukham	2

Q	Answer	Marks
	(comfortable) + āsanam (posture). “Posture should be steady and comfortable.”	
2	BOTH steady AND comfortable. Not one or the other.	2
3	prayatna-śaithilya = relaxation of effort. ananta-samāpatti = absorption into the infinite.	2
4	When āsana is steady-and-comfortable, prāṇāyāma is the regulation (cessation of the flow) of inhalation and exhalation. The verse explicitly says āsana comes first.	2
5	Forceful retention can cause dizziness, hyperventilation, or other harm without trained supervision and contraindication screening.	2

Total: 10.

Module 8 (Senior) — Summative Quiz

Time: 35 min · **Marks:** 35 · **Closed-book**

Name: _____ **Class:** _____

Section A — Multiple choice (5 × 1 = 5 marks)

1. Patañjali’s definition of yoga (YS 1.2) is: a) Skill in action b) Settling of the fluctuations of consciousness c) Union of body, breath, and mind d) Devotion to Īśvara
2. *sthira-sukham āsanam* (YS 2.46) means: a) Posture is a spiritual practice b) Posture should be steady and comfortable c) Posture should be held until pain d) Posture is the third limb
3. Modern Sūrya Namaskāra in its 12-pose form was codified: a) In Patañjali’s Yoga Sūtras (~2nd c. BCE) b) In the Bhagavad-gītā c) In the early 20th century (Raja of Aundh, 1928) d) In Iyengar’s *Light on Yoga* (1966)
4. Polyvagal theory (Porges) proposes how many tiers of autonomic response? a) 2 b) 3 c) 4 d) 5
5. Which scholar argues that modern postural yoga is largely a 20th-century synthesis? a) Iyengar b) Patañjali c) Singleton d) Bryant

Section B — Primary source (6 marks)

6. (3 marks) Translate word-by-word: *sthira-sukham āsanam* (YS 2.46).

7. (3 marks) State what YS 2.47 adds to YS 2.46. Use the Sanskrit terms *prayatna-śaithilya* and *ananta-samāpatti* in your answer.

Section C — Eight limbs (4 marks)

8. Write all 8 limbs in order, with English meanings.

Section D — Physiology (4 marks)

9. (2 marks) Why does slow exhalation tend to shift the body toward calm?

10. (2 marks) Name one limit or critique of polyvagal theory.

Section E — History (4 marks)

11. (2 marks) Name 2 specific claims Singleton (2010) makes about the 20th-century history of postural yoga.

12. (2 marks) What does Singleton NOT claim? Name 2 things commonly misattributed to his thesis.

Section F — Open response (12 marks)

13. In 1–2 pages, argue your position on the following:

“Knowing the 20th-century history of modern postural yoga (Singleton 2010) diminishes its value for practitioners today.”

Required: - Clear position - ≥1 citation - A steel-manned counter-argument

Answer key (summary)

Q	Answer	Marks
1	b	1
2	b	1
3	c	1
4	b (three)	1

Q	Answer	Marks
5	c	1
6	sthira = steady, sukham = comfortable, āsanam = posture.	3
7	YS 2.47 says the dual condition is achieved through prayatna-śaithilya (relaxation of effort) and ananta-samāpatti (absorption into the infinite).	3
8	yama, niyama, āsana, prāṇāyāma, pratyāhāra, dhāraṇā, dhyāna, samādhi (with meanings)	4
9	Slow exhale stimulates the vagus nerve → parasympathetic activation → calming.	2
10	Examples: Grossman & Taylor (2007) dispute specific vagal claims; the three-tier model may be overspecified vs. simpler models.	2
11	Examples: 12-pose SN is early-20th c.; modern postural yoga draws on European physical-culture as well as Indian sources; the international spread is via Krishnamacharya's students.	2
12	Singleton does NOT claim: (a) yoga is fake; (b) Patañjali is recent; (c) Indians didn't invent yoga. (Any 2.)	2
13	Marked on clear position (4), citation (2), counter-argument (3), avoidance of false dichotomy (3).	12

Total: 35.

Homework

Homework — Module 8 (Senior)

Night	Task	Time
Day 1 → 2	Read Bryant's intro to <i>Yoga Sūtras</i> . Note 1 question.	25 min
Day 2 → 3	Memorise the 8 limbs in Sanskrit.	15 min

Night	Task	Time
Day 3 → 4	Practise tādāsana + vṛkṣāsana + balāsana. Reflect: <i>sthira-sukham</i> ?	15 min
Day 4 → 5	Read YS 2.49–53 in Bryant’s translation.	25 min
Day 5 → 6	Prepare for Porges reading; read teacher-prepared synopsis.	15 min
Day 6 → 7	Read Singleton’s intro to <i>Yoga Body</i> .	30 min
Day 7 → 8	Submit project plan: topic + thesis + 3 sources.	30 min
Day 8 → 9	Draft 1.5–2 pages.	60–90 min
Day 9 → 10	Complete draft, format, citations.	60–90 min

Writing Prompts

Writing Prompts — Module 8 Yoga (Senior)

Reflection

- (Day 1)** Before this module, what did “yoga” mean to you? Which of the three definitions (Patañjali / Gītā / modern) most matched your assumption?
- (Day 3)** Does YOUR yoga practice meet Patañjali’s *sthira-sukham* criterion? Be honest.

Explanatory

- (Day 5)** Explain what YS 2.49 says about the relationship between āsana and prāṇāyāma in 5 sentences.
- (Day 6)** Explain polyvagal theory to a Grade 9 reader in 1 page.

Argumentative

- (Day 7)** Argue: “Singleton’s (2010) historical thesis enriches rather than diminishes yoga practice.” 1–2 pages.
- (Day 7)** Argue the OPPOSITE: “Singleton’s framing is a Western-academic project that doesn’t change what yoga IS for the practitioner.”

Cross-disciplinary

7. (Stretch) Compare Patañjali's *sthira-sukham* with a similar dual condition from another tradition (Japanese *seiza* meditation? Christian centering prayer?). What's the philosophical move both make?

8. (Stretch) Read a contemporary peer-reviewed study on yoga's effects (Pascoe 2017 meta-analysis recommended). Critique the methodology in 1 page.

Grading

Same as Module 2/5 Senior — thesis, citations, counter-argument, prose.

Activity 01 — Testing Patañjali's Dual Criterion

Module: Yoga (Senior) · **Day:** 3 · **Time:** 15 min within Day 3

Goal

Students hold three postures and self-assess against YS 2.46 (*sthira-sukham āsanam*).

Materials

- Yoga mats
- Stopwatch
- Reflection worksheet (below)

Instructions

Hold each pose. After each, immediately rate: - **Sthira** (steady): 1–5 - **Sukha** (comfortable): 1–5 - Both?

Sequence

Pose	Hold	Notes
Tāḍāsana	60 sec	Standing, easiest to score
Vṛkṣāsana (right)	30 sec	Balance challenge
Balāsana	90 sec	Rest — should be easy on both

Reflection worksheet

Pose	Sthira (1–5)	Sukha (1–5)	Both? Y/N	Notes
Tāḍāsana				
Vṛkṣāsana				

Pose	Sthira (1–5)	Sukha (1–5)	Both? Y/N	Notes
Balāsana				

Discussion (5 min after)

- Which pose best met BOTH criteria for you, today?
- Which one was steady but not comfortable? Comfortable but not steady?
- What might *prayatna-śaithilya* (relaxation of effort) look like for the pose where you struggled most?

Safety

- Standard rules: opt-out anytime, vṛkṣāsana foot above/below knee not on it.

Differentiation

- **Tier up:** Try adho-mukha-śvānāsana (downward dog) — does it meet the criterion?
- **Tier down:** Skip vṛkṣāsana; do tāḍāsana + balāsana only.

Clean-up

- Worksheets collected — feeds into Day 3’s reflection writing.

Project Brief

Project — Module 8 Yoga (Senior)

Type: 5–6 page research-synthesis paper · **Marks:** 65

Topic options

Choose one or propose:

1. **Does the counted-breath prāṇāyāma practice we learned have measurable effects on heart-rate variability?** Review 2–3 contemporary studies and reach a conclusion.
2. **Patañjali defines āsana as “steady and comfortable” (YS 2.46). Does modern postural yoga meet this criterion?** Use 3 specific postures as evidence.
3. **Evaluate Singleton’s (2010) historical thesis** on modern postural yoga. Argue for, against, or qualified position.
4. **Compare Patañjali’s prāṇāyāma framework (YS 2.49–53) with the polyvagal theory’s account** of breath effects. What does each frame capture?
5. **Trace one specific āsana** from its earliest textual mention (if any) to its modern form. What does the history reveal?

Required elements

Your paper **MUST** cite:

- **One** classical primary source (YS, BG, or SB verse in IAST + translation).
- **One** historical scholarly source (Singleton, De Michelis, Sjoman, or equivalent).
- **One** peer-reviewed modern research paper (Porges, Streeter, Pascoe, or your own discovery).

Plus: - Clear thesis in opening paragraph. - One steel-manned counter-argument with response. - Works-cited page (MLA or Chicago).

Format

- 5–6 pages, double-spaced, 12pt.
- Title page with name, class, thesis.

Schedule

- Day 7: choose topic.
- Day 8: outline submitted.
- Day 9: draft (≥ 2 pages).
- Day 10: paper submitted.

Rubric

See `rubric.md`.

Self-reflection (1 paragraph)

1. Topic + why.
2. Most challenging part.
3. With 3 more days, what to research deeper.

Rubric

Paper Rubric — Module 8 (Senior)

Paper marks: 65 (with summative 35 → 100 total).

Criteria

Criterion	4	3	2	1	Weight
Thesis clarity	Crystal-clear, specific,	Clear	Vague	None	× 3

Criterion	4	3	2	1	Weight
Classical source	defensible YS or BG verse quoted in IAST + translation + interpreted	Cites verse, interprets adequately	Vague	None	× 2
Historical source	Singlet on or De Michel is cited substantively	Cites scholarship	Vague	None	× 2
Modern research	Peer-reviewed study cited and critiqued	Study cited	Vague reference	None	× 2
Counter-argument	Steelman d opposing view + reasoned response	Counter present	Acknowledged	Straw manne d	× 2
Synthesis quality	Integrates 3 source types into cohere	Mostly integrates	Partly integrates	Cherry -picks one	× 2

Criterion	4	3	2	1	Weight
	nt argum ent				
Avoidance of false dichotomy	Holds “yoga is valuab le” AND “histor y is what it is”	Mostly nuance d	Hedge s	Reduc es to either- or	× 1
Prose / structure	Polish ed	Clear	Some issues	Hard to follow	× 1

Max raw: 60. Paper = (sum × weights) × 65/60.

Format deductions

- 5–6 pages (deduct 5 outside)
- Title page + thesis (deduct 5 missing)
- Works-cited (deduct 5 missing/wrong)
- IAST diacritics correct (deduct 3)

Safety violation in project recommendations

- If a paper recommends an unsafe practice without contraindication note, deduct 5 from “Conceptual accuracy / safety.”